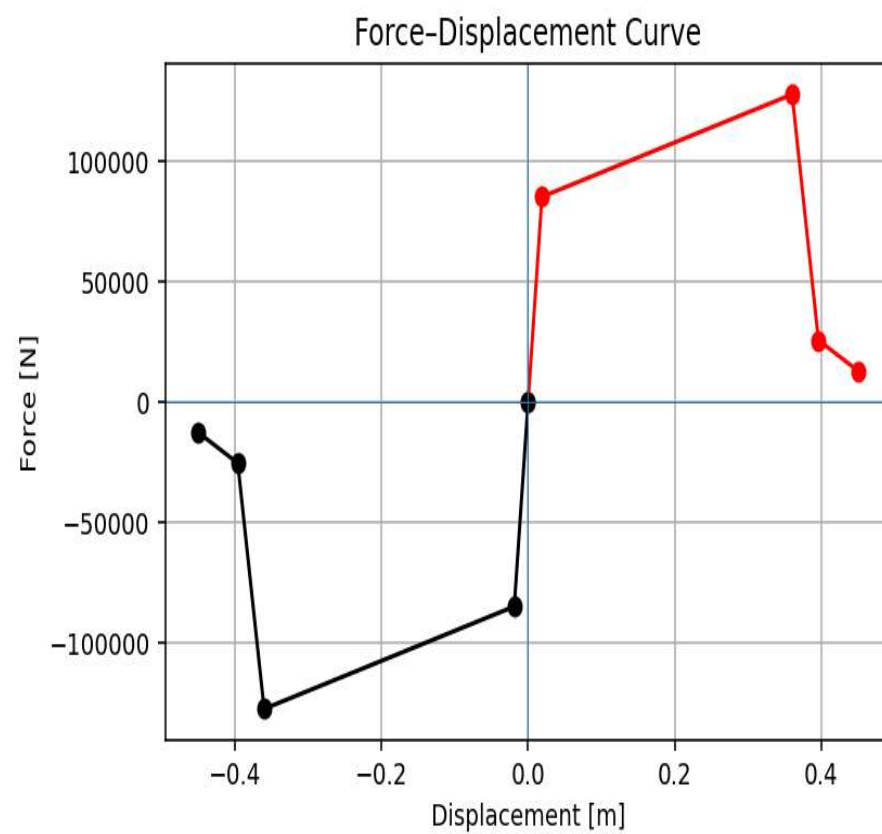
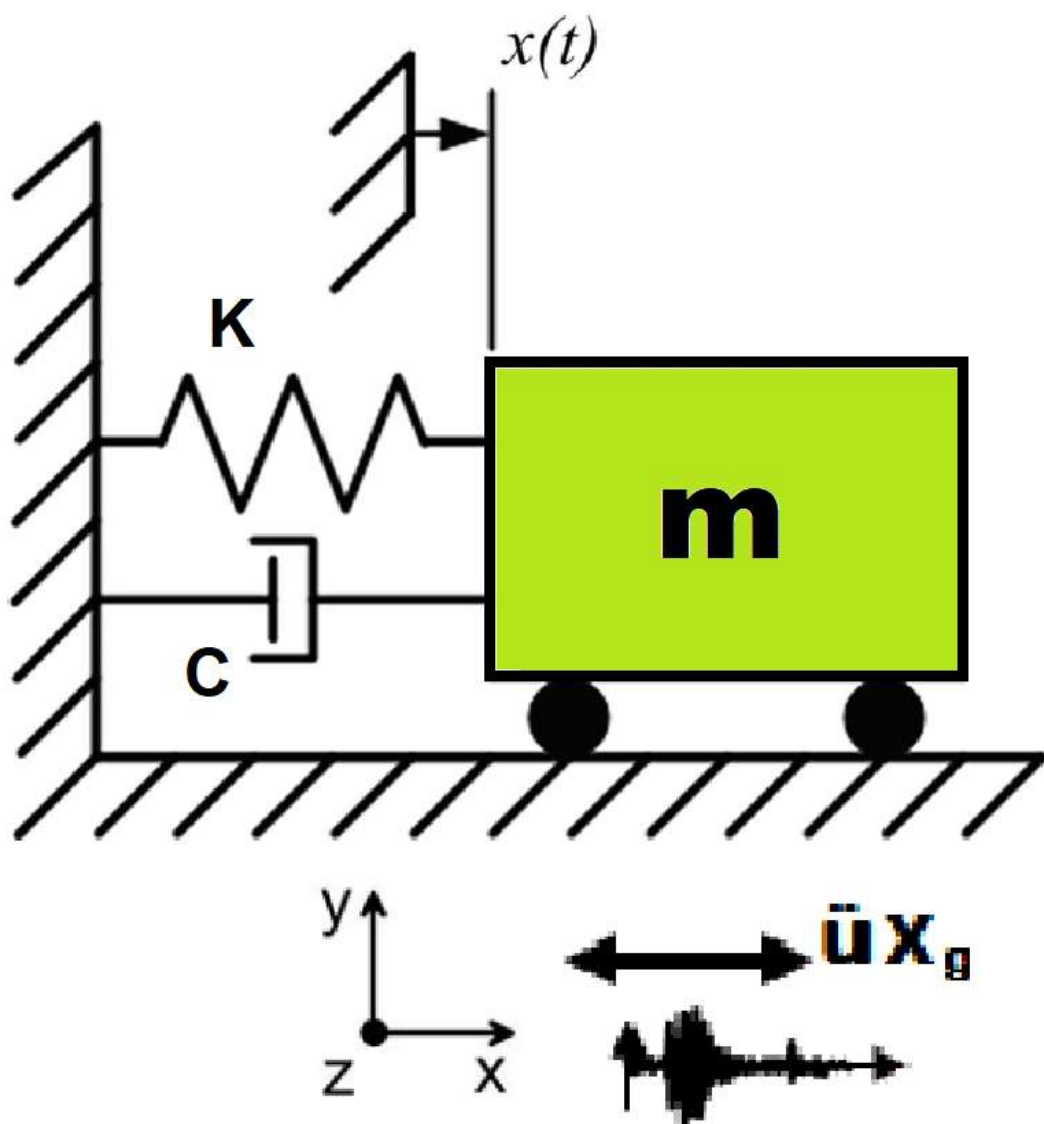


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**FRAGILITY ANALYSIS BASED ON
ACCELERATION AND STRUCTURAL DUCTILITY
DAMAGE INDEX AND EVALUATION OF ENERGY
DISSIPATION CAPACITY INDEX WITH
INCREMENTAL DYNAMIC ANALYSIS (IDA) OF A
SINGLE-DEGREE-OF-FREEDOM (SDOF) SYSTEM
UTILIZING 30 GROUND MOTIONS IN OPENSEES**

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

Δ_u = Lateral Ultimate Displacement from Pushover Analysis

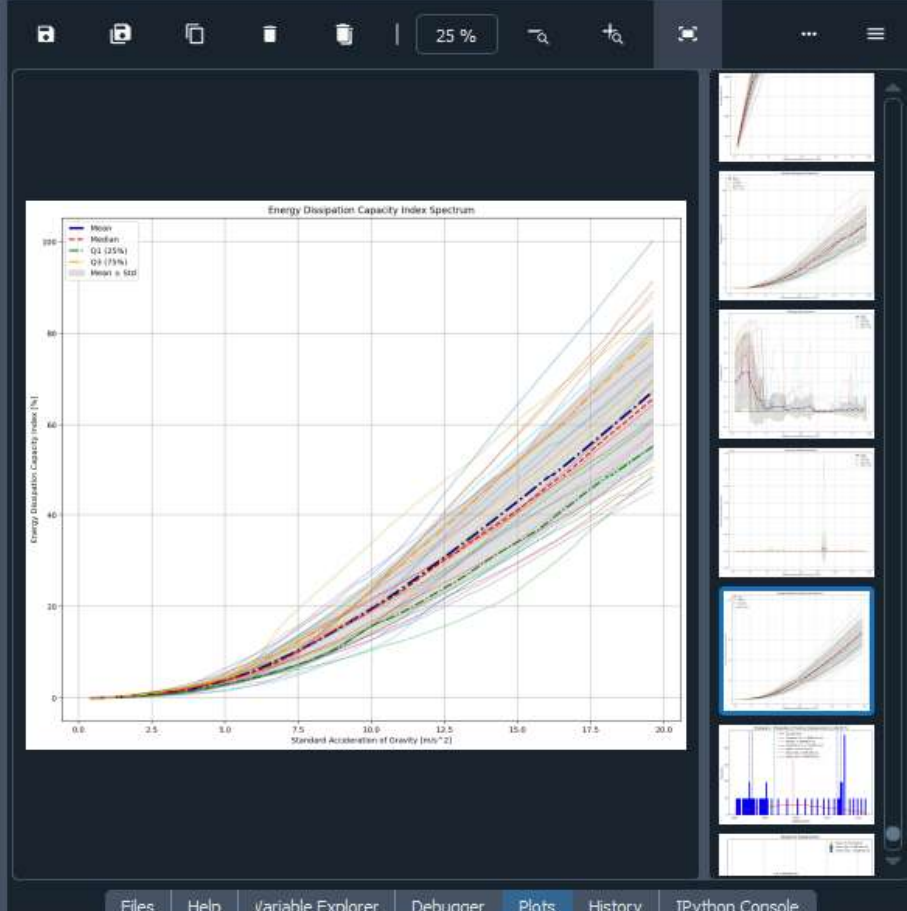
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EX+\SDOF_IDA_ENERGY DISSIPATION CAPACITY INDEX.py

SDOF_IDA_ENERGY DI...CAPACITY INDEX.py x FRAGILITY_CURVE_FUN.py x

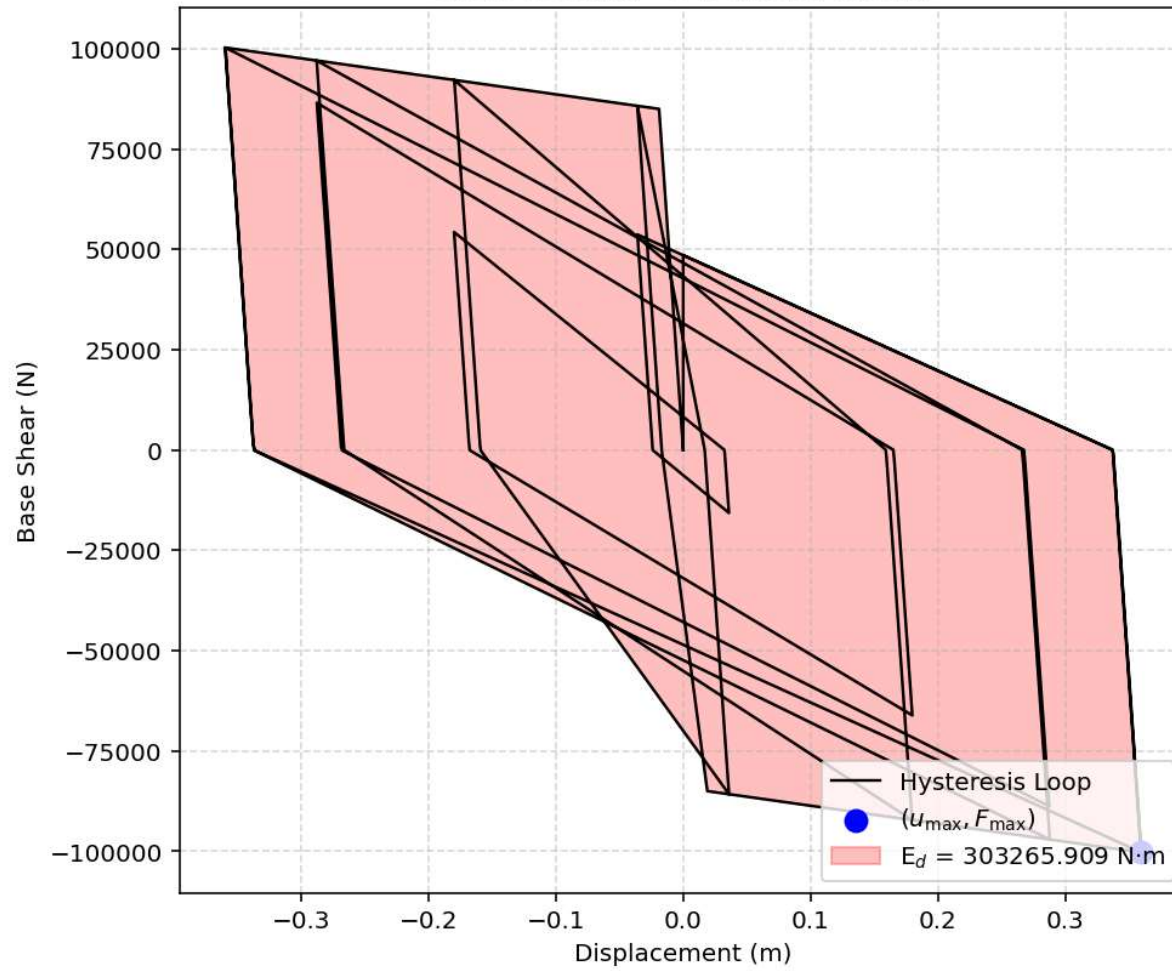
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1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # FRAGILITY ANALYSIS BASED ON ACCELERATION AND STRUCTURAL DUCTILITY DAMAGE INDEX AND EVAL
4 # OF ENERGY DISSIPATION CAPACITY INDEX WITH INCREMENTAL DYNAMIC ANALYSIS (IDA)
5 # OF A SINGLE-DEGREE-OF-FREEDOM (SDOF) SYSTEM UTILIZING 30 GROUND MOTIONS IN
6 #
7 # This program performs Incremental Dynamic Analysis (IDA) on a Single-Degree-of-Freedom (SDO
8 # subjected to 30 seismic ground motions. The analysis evaluates the structural response unde
9 # levels of seismic intensity.
10 # The framework is designed to support researchers and engineers in assessing the probabilist
11 # performance of structures, with a focus on understanding the impact of uncertainty on struc
12 # response and design.
13 #
14 # Key Features:
15 # - Simulation of SDOF system using OpenSees.
16 # - Incremental scaling of ground motions for IDA.
17 # - Probabilistic fragility assessment based on predefined damage states.
18 # - Visualization of structural response and fragility curves.
19 # - Export of results for further analysis.
20 #
21 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAIEI (QASHQAI)
22 # EMAIL: salar.d.ghashghaei@gmail.com
23 #####
24
25 """
26 This code implements a comprehensive nonlinear incremental dynamic analysis framework for
27 performance-based earthquake engineering assessment of single-degree-of-freedom
28 (SDOF) systems. The methodology combines traditional nonlinear time-history
29 analysis with modern probabilistic and machine learning techniques for advanced
30 structural performance evaluation.
31
32 KEY ENGINEERING OBJECTIVES:
33 1. Comparative assessment of hysteretic models for seismic response prediction

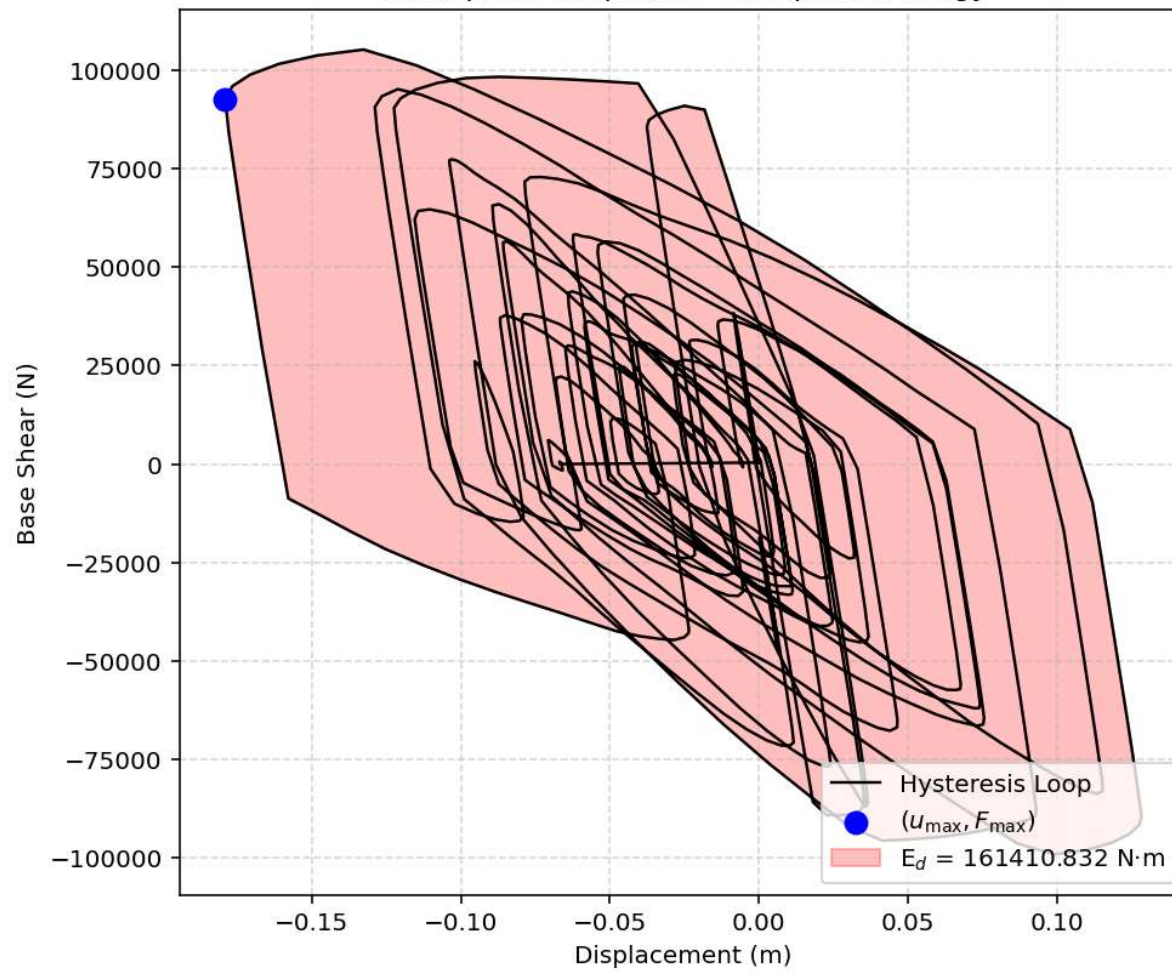
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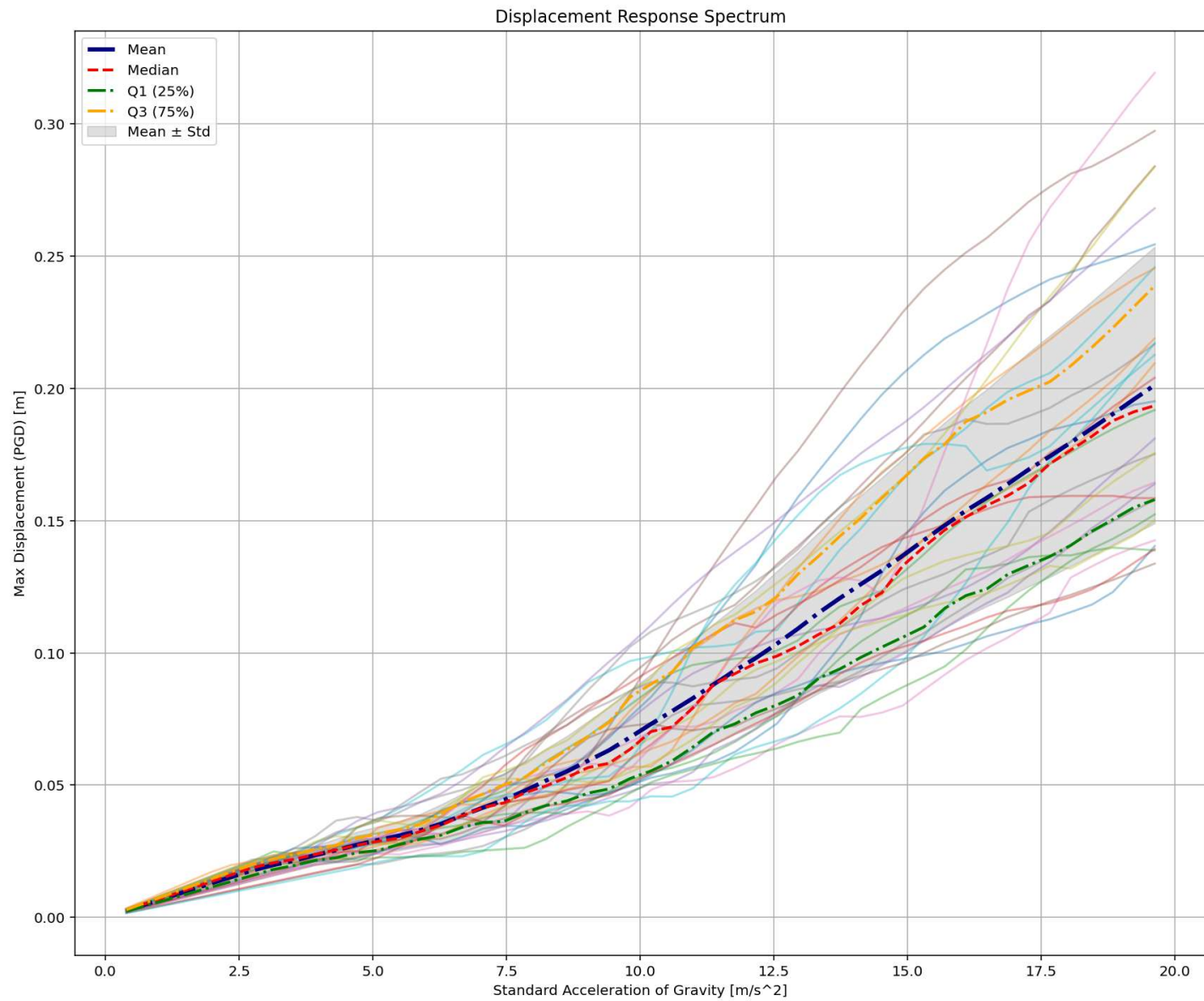


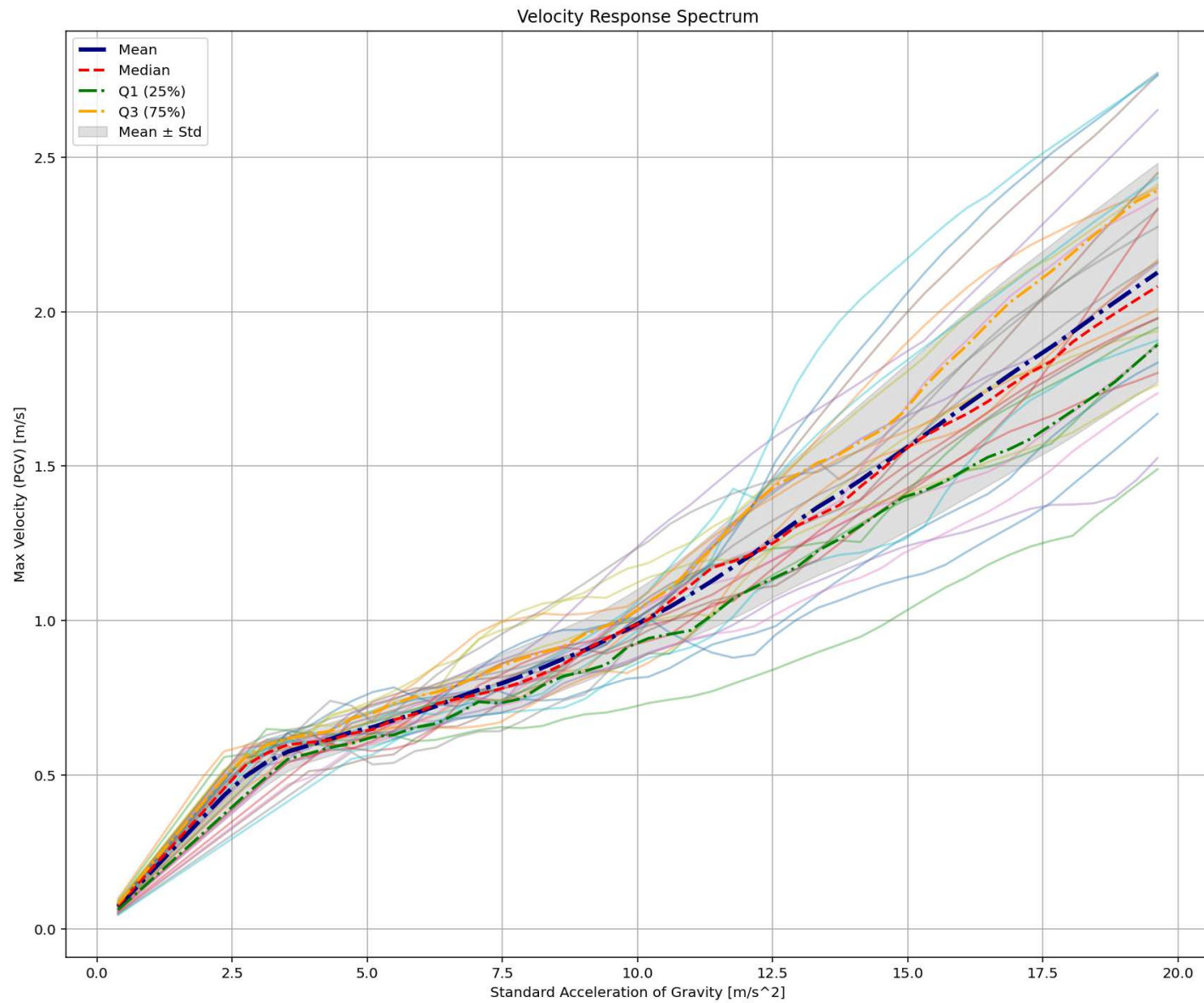
Cyclic Loading - Dissipated Energy

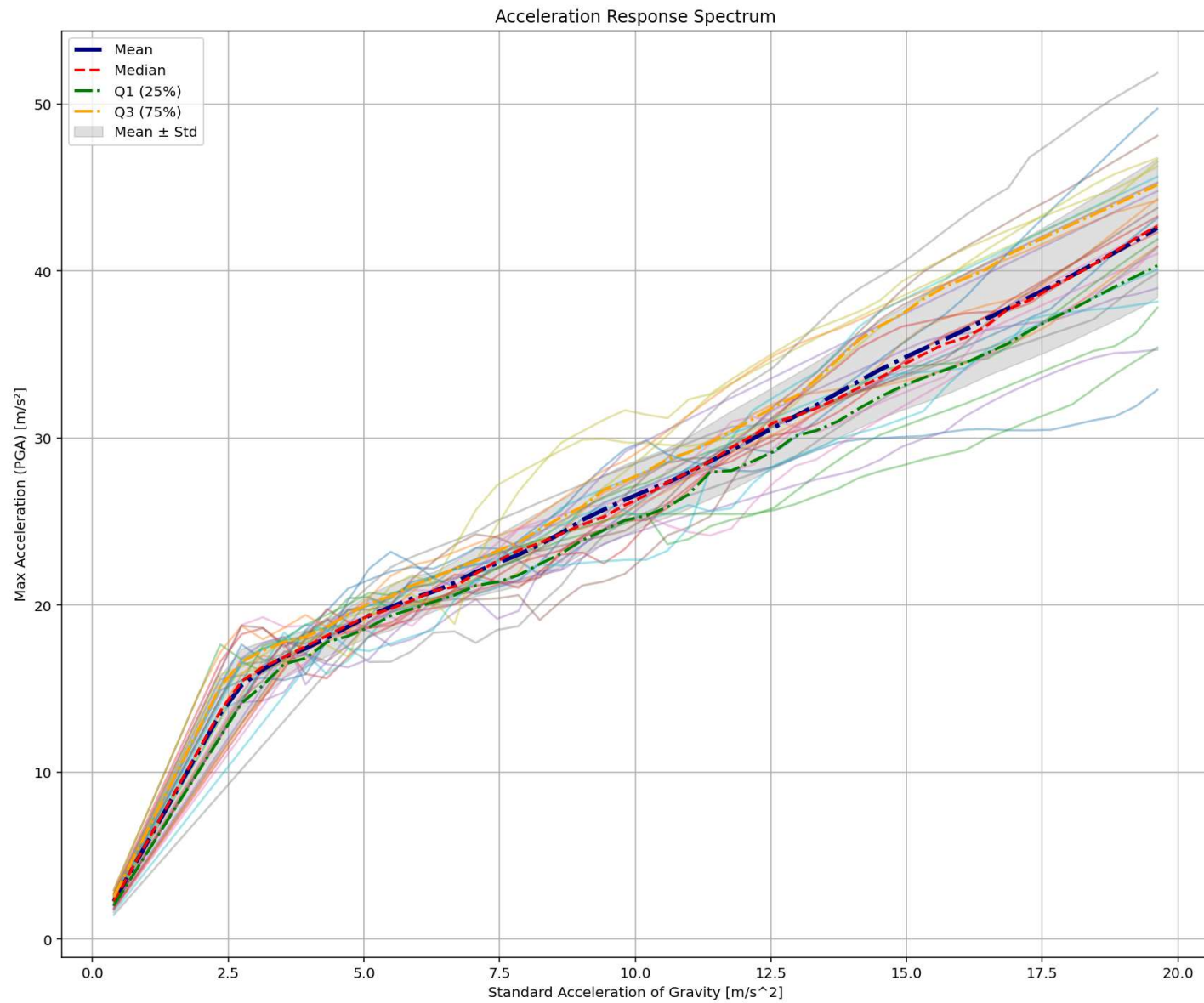


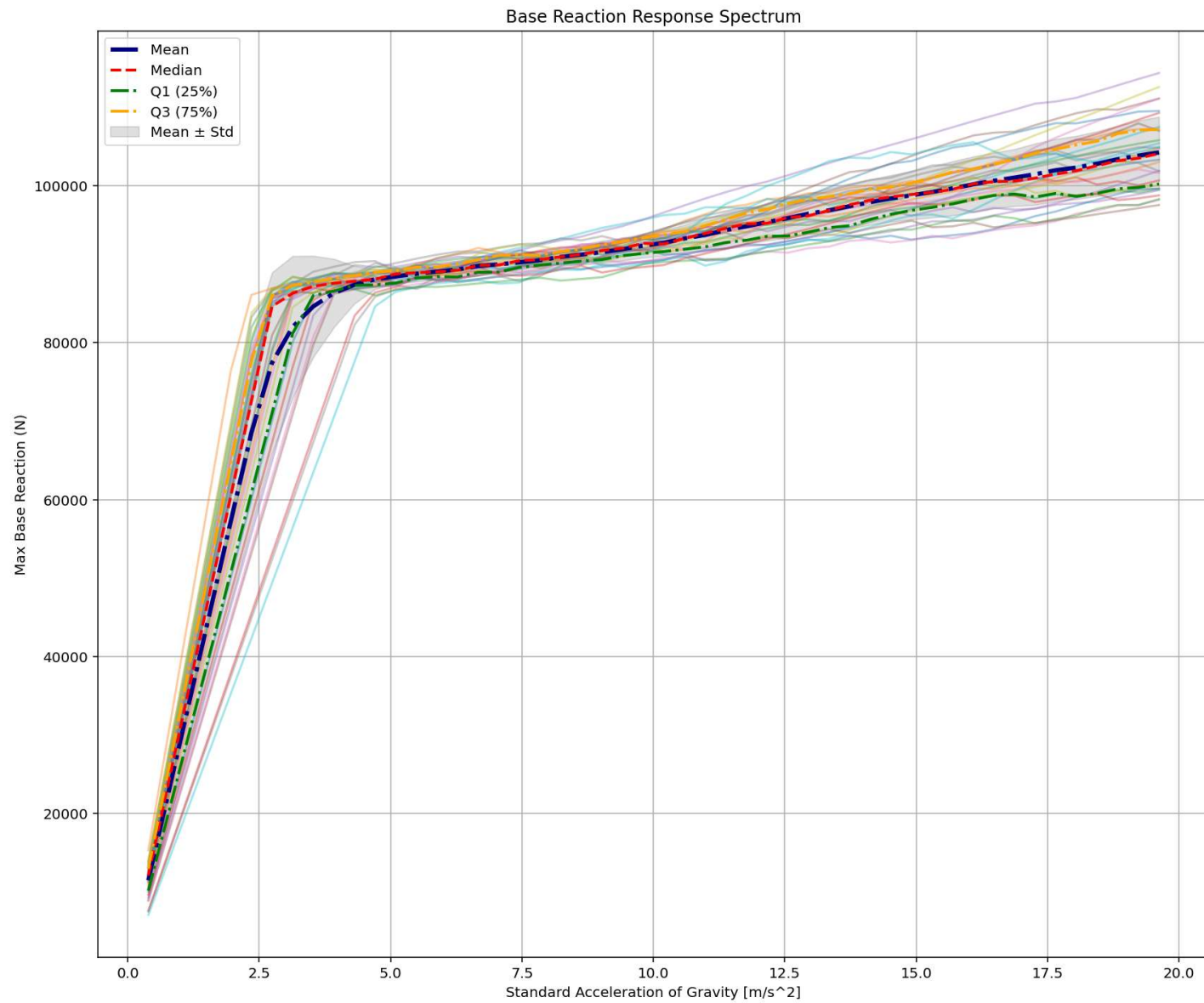
Earthquake Response - Dissipated Energy

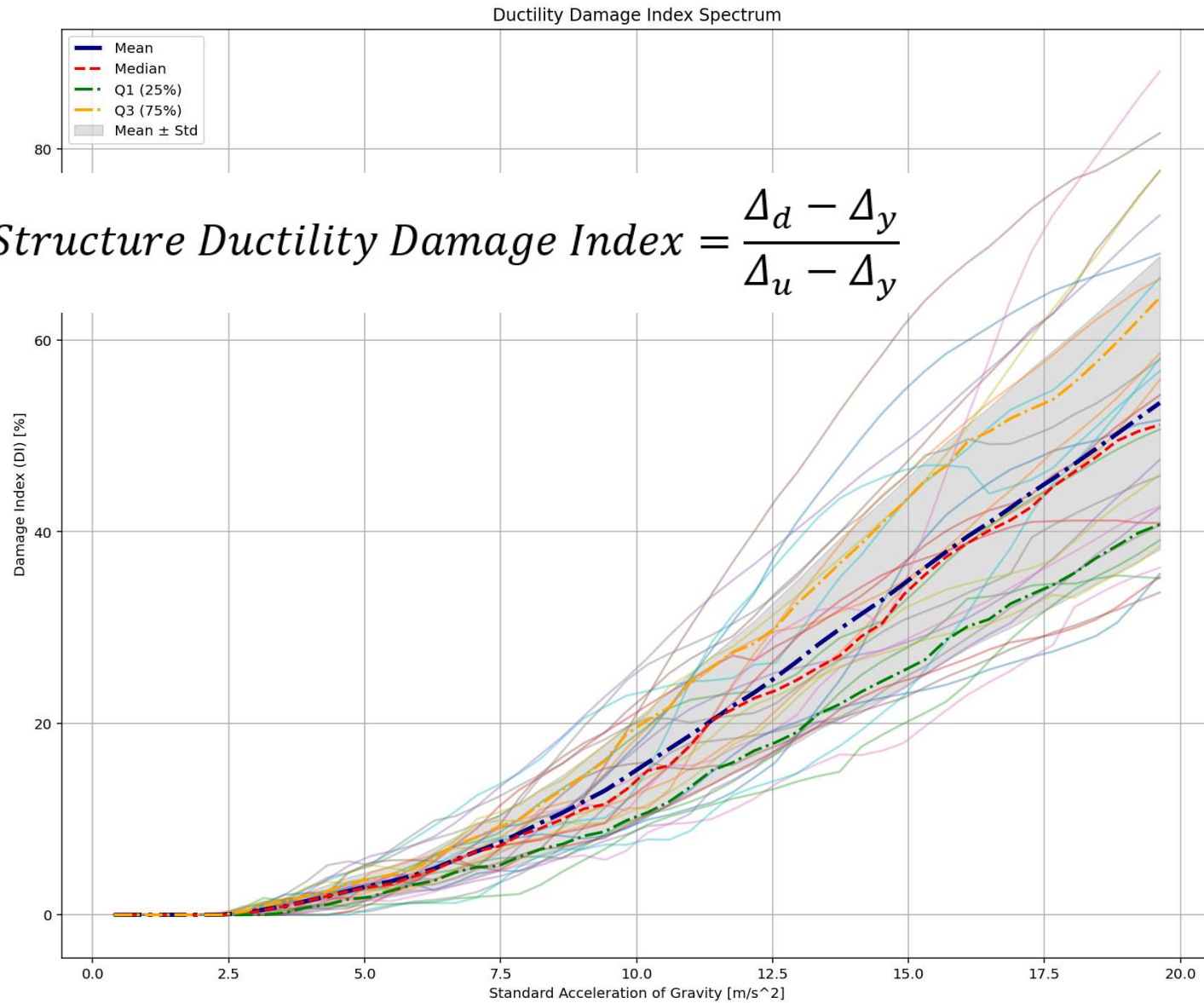


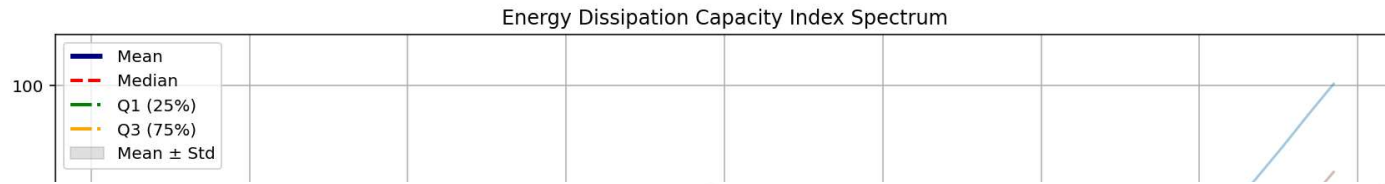




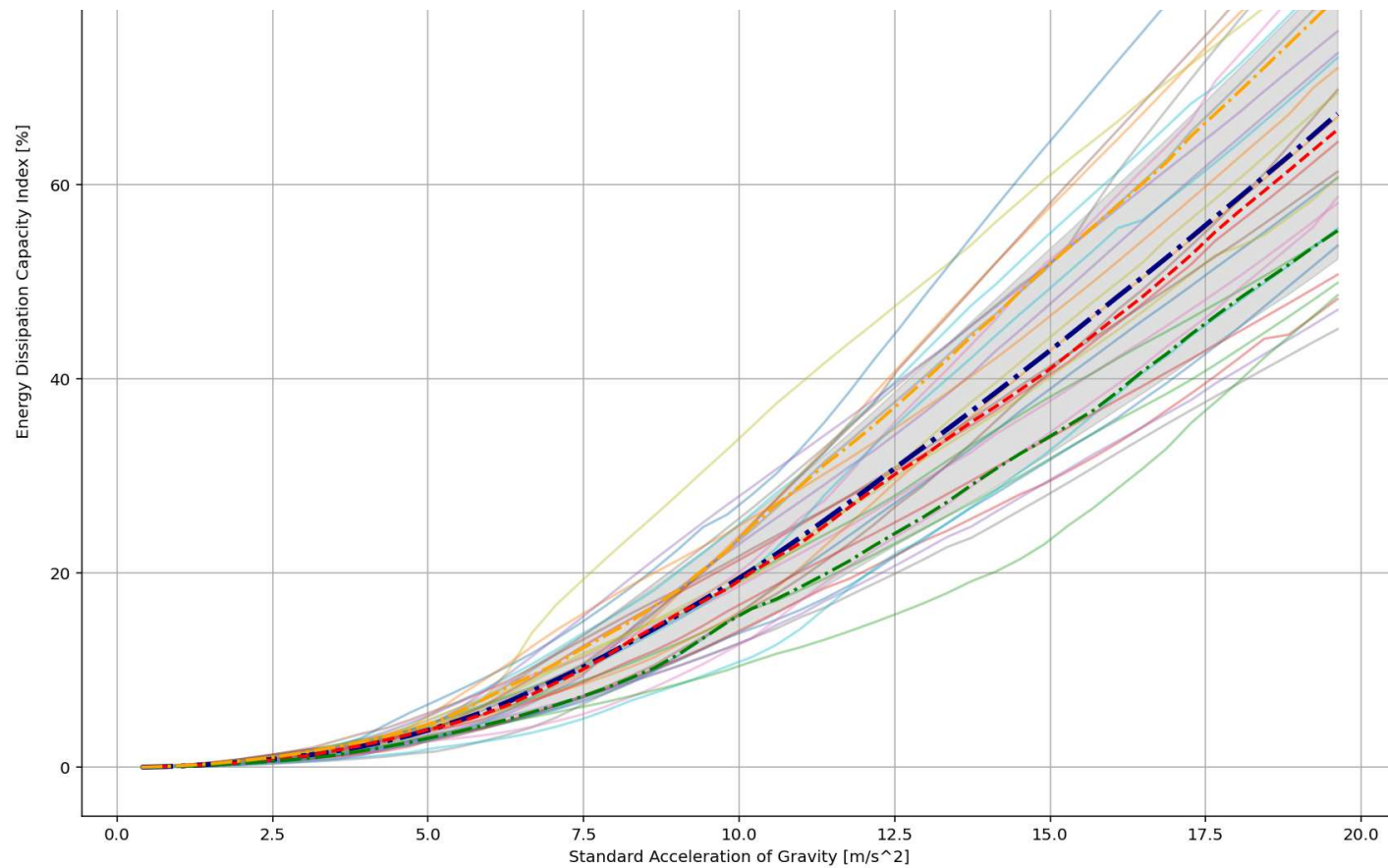




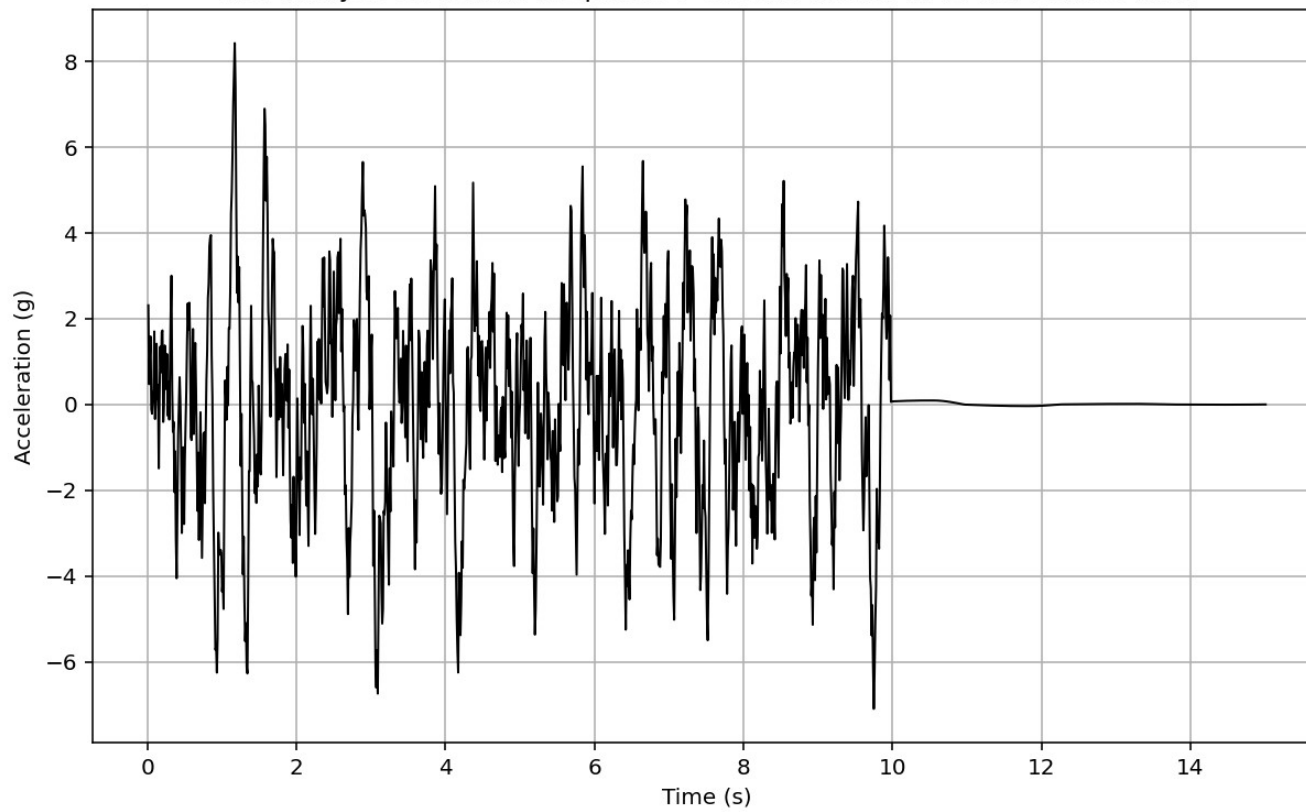


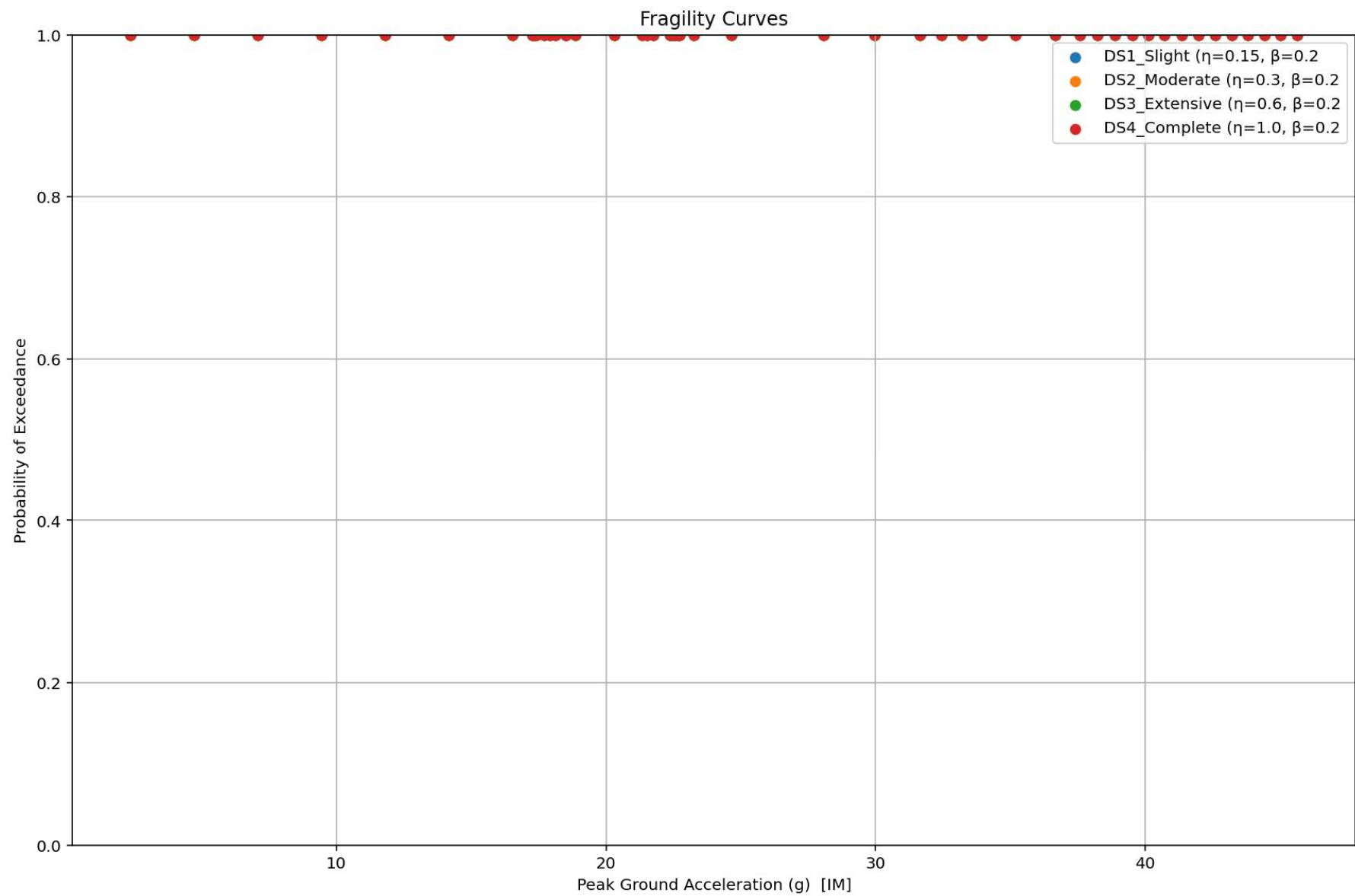


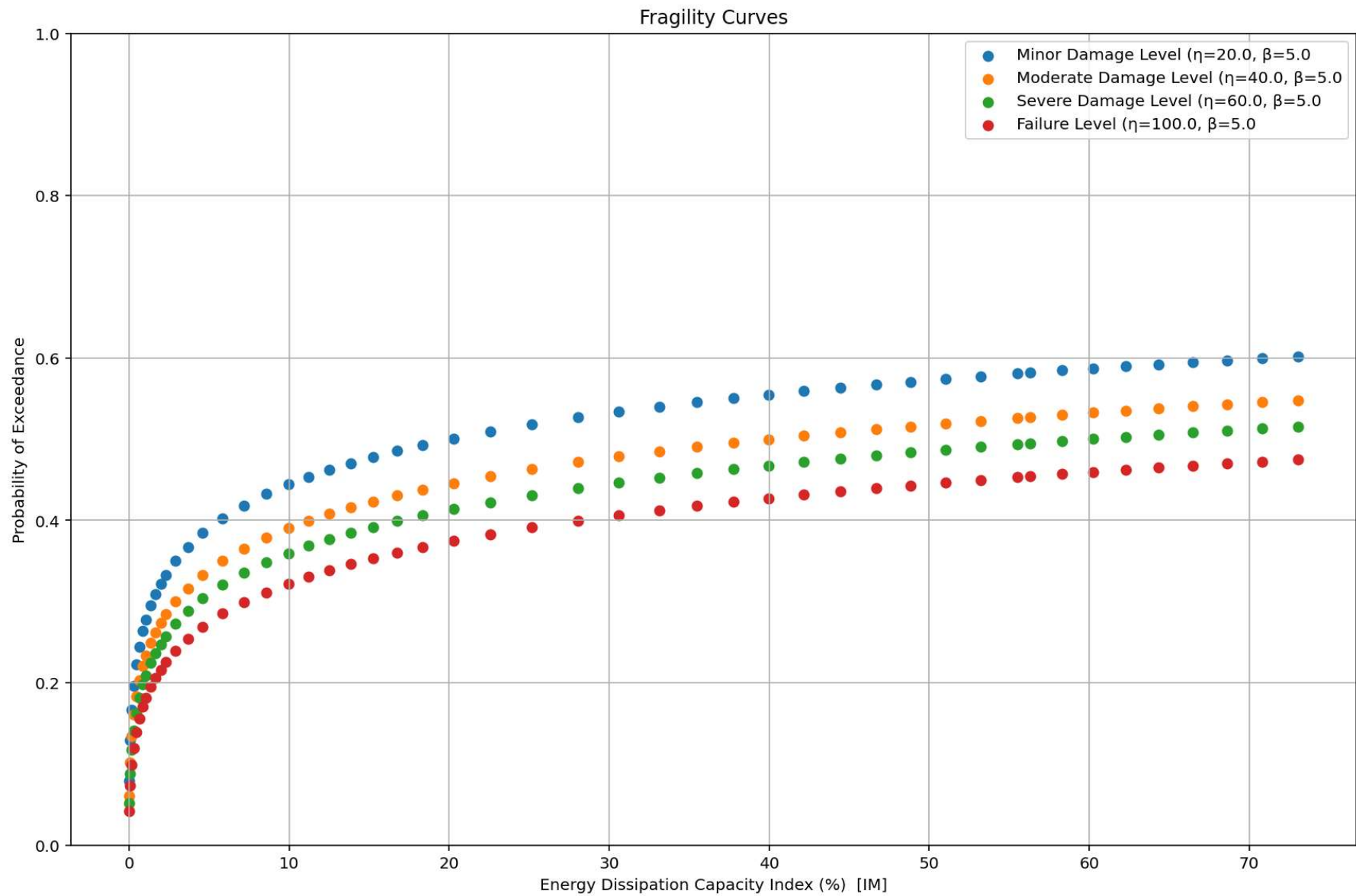
$$\text{Energy Dissipation Capacity Index} = \frac{E_d(\text{Dissipated Energy from Earthquake})}{E_d(\text{Dissipated Energy from Cyclic Displacement})} * 100$$



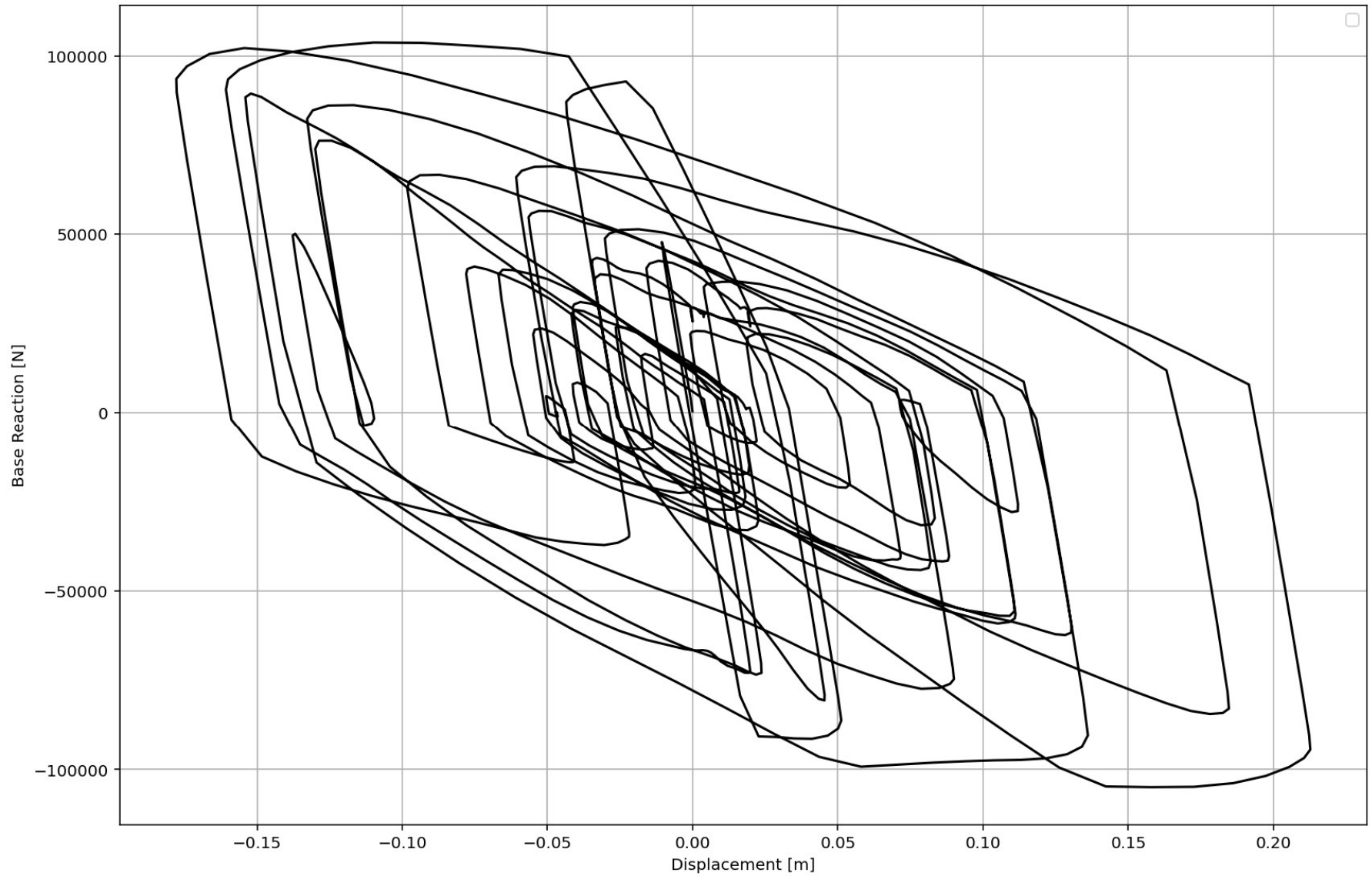
Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 8.4259

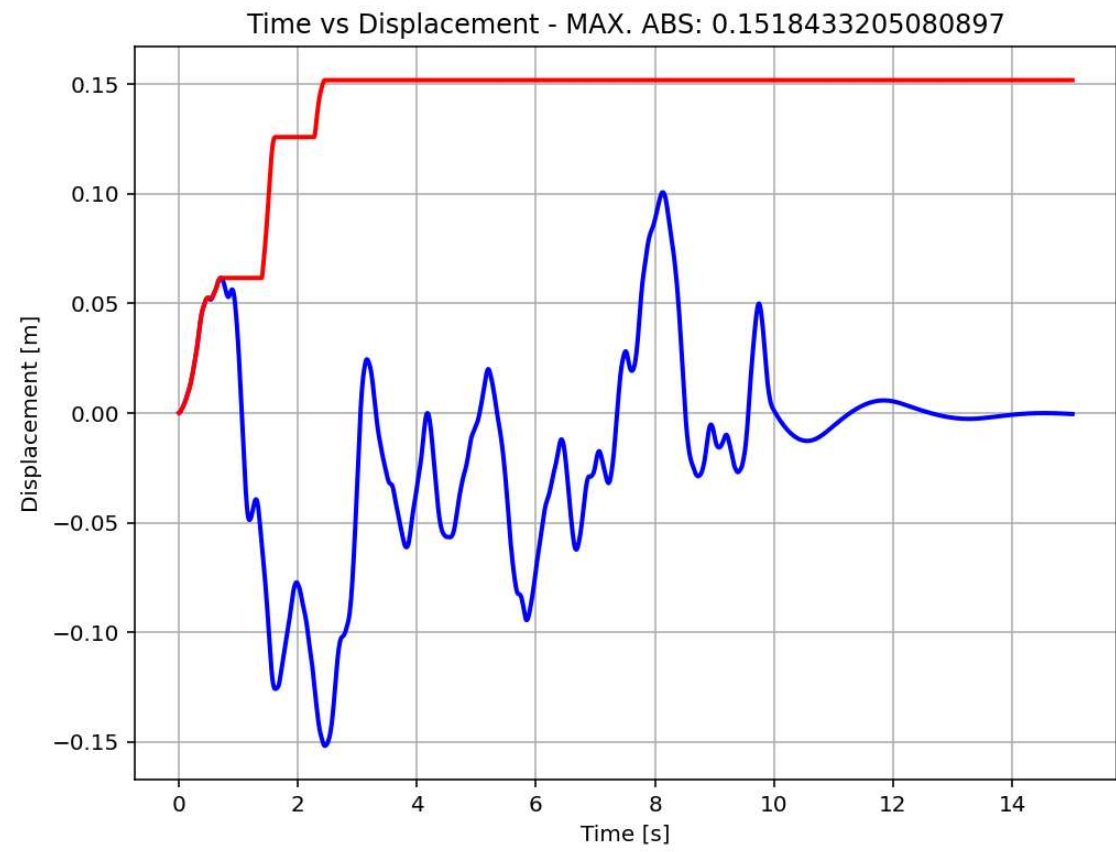


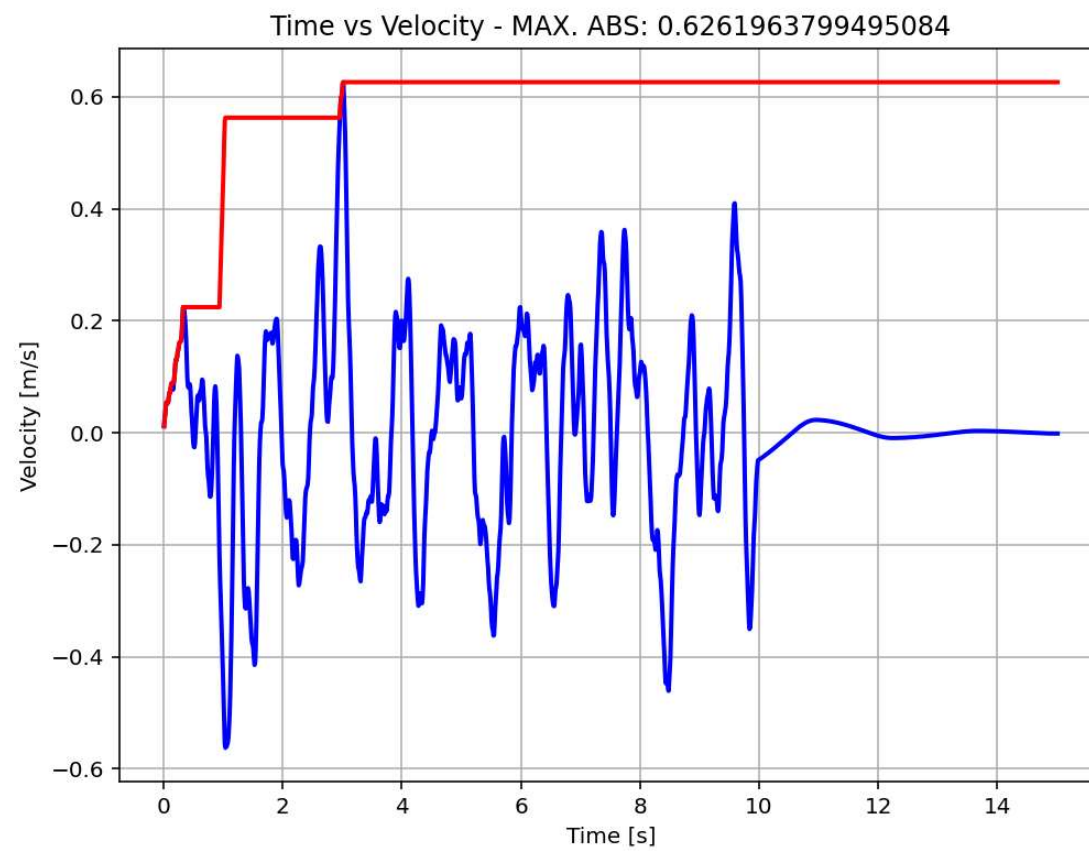


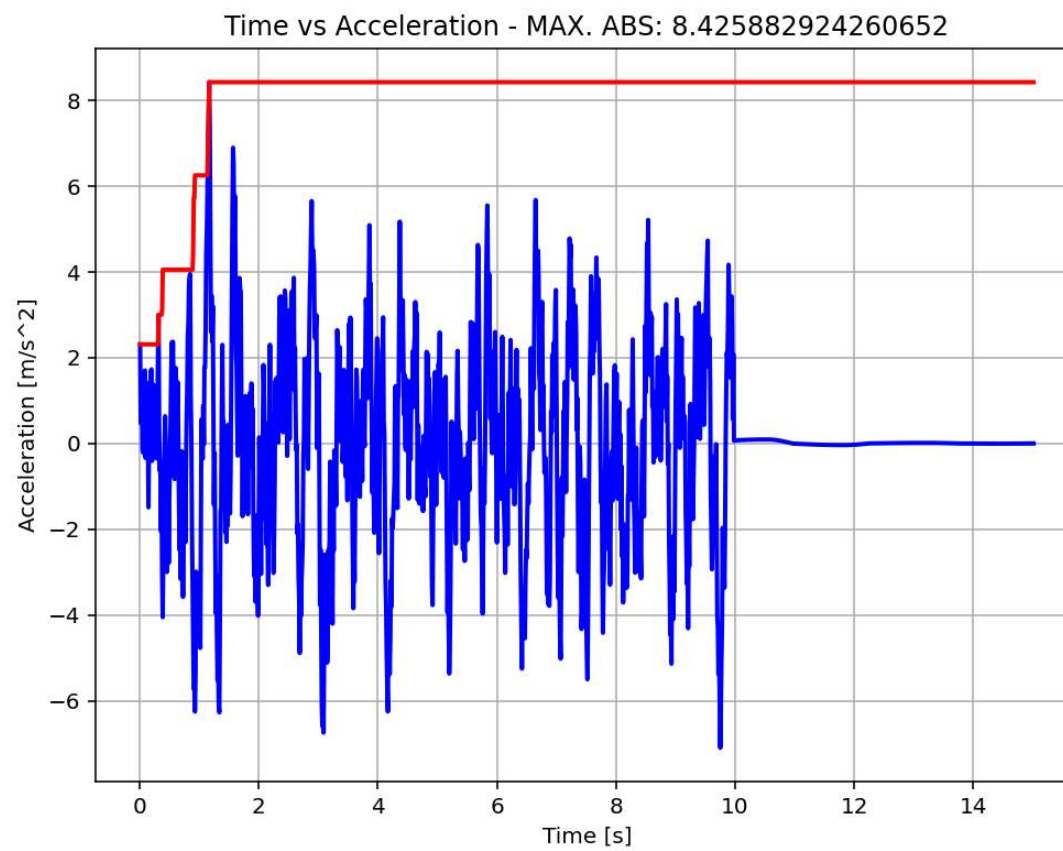


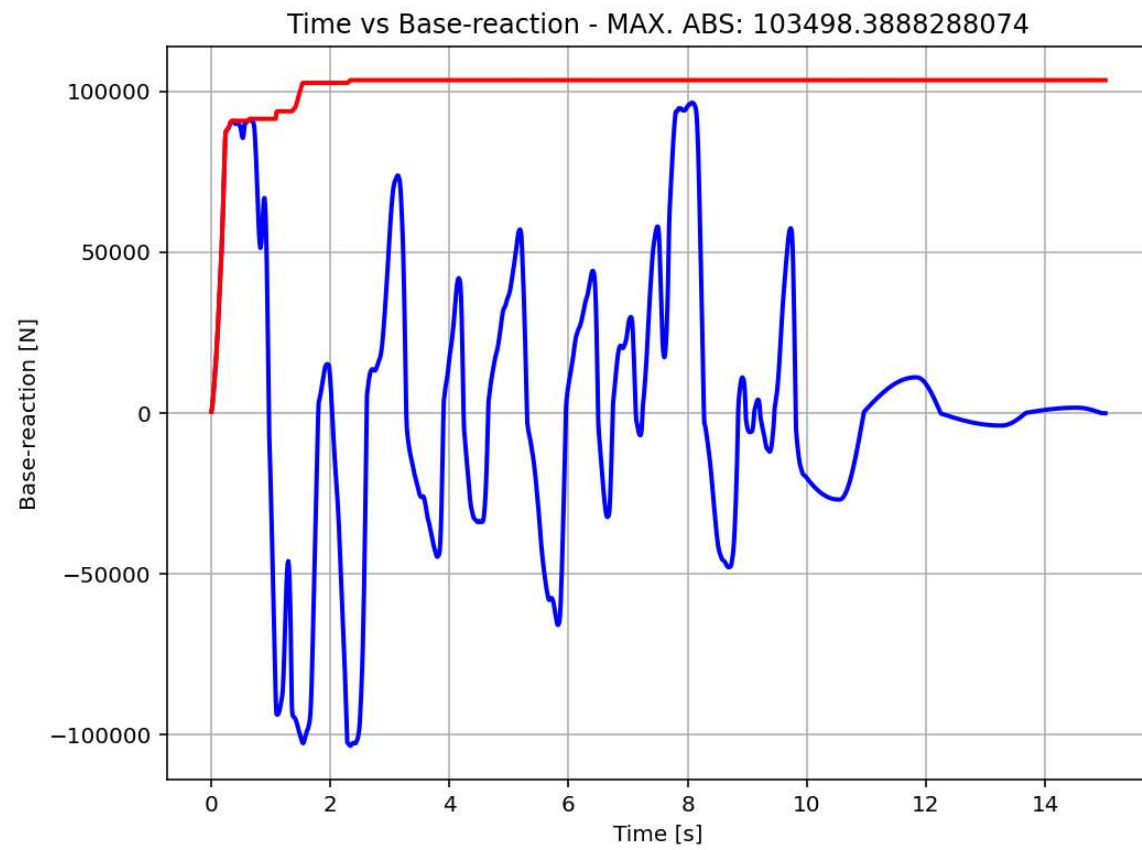
Displacement & Base Reaction Relation From Last Dynamic Analysis

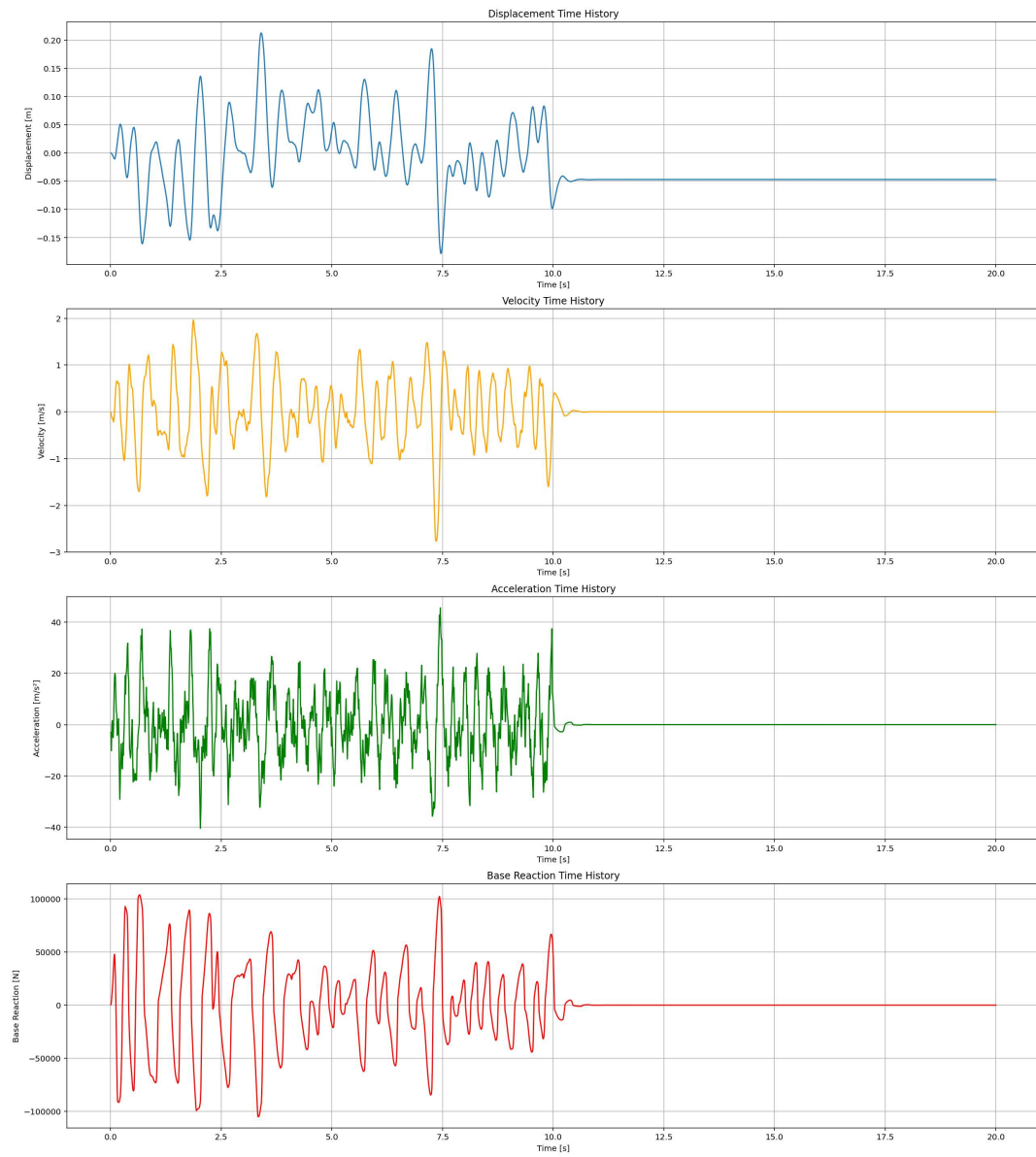




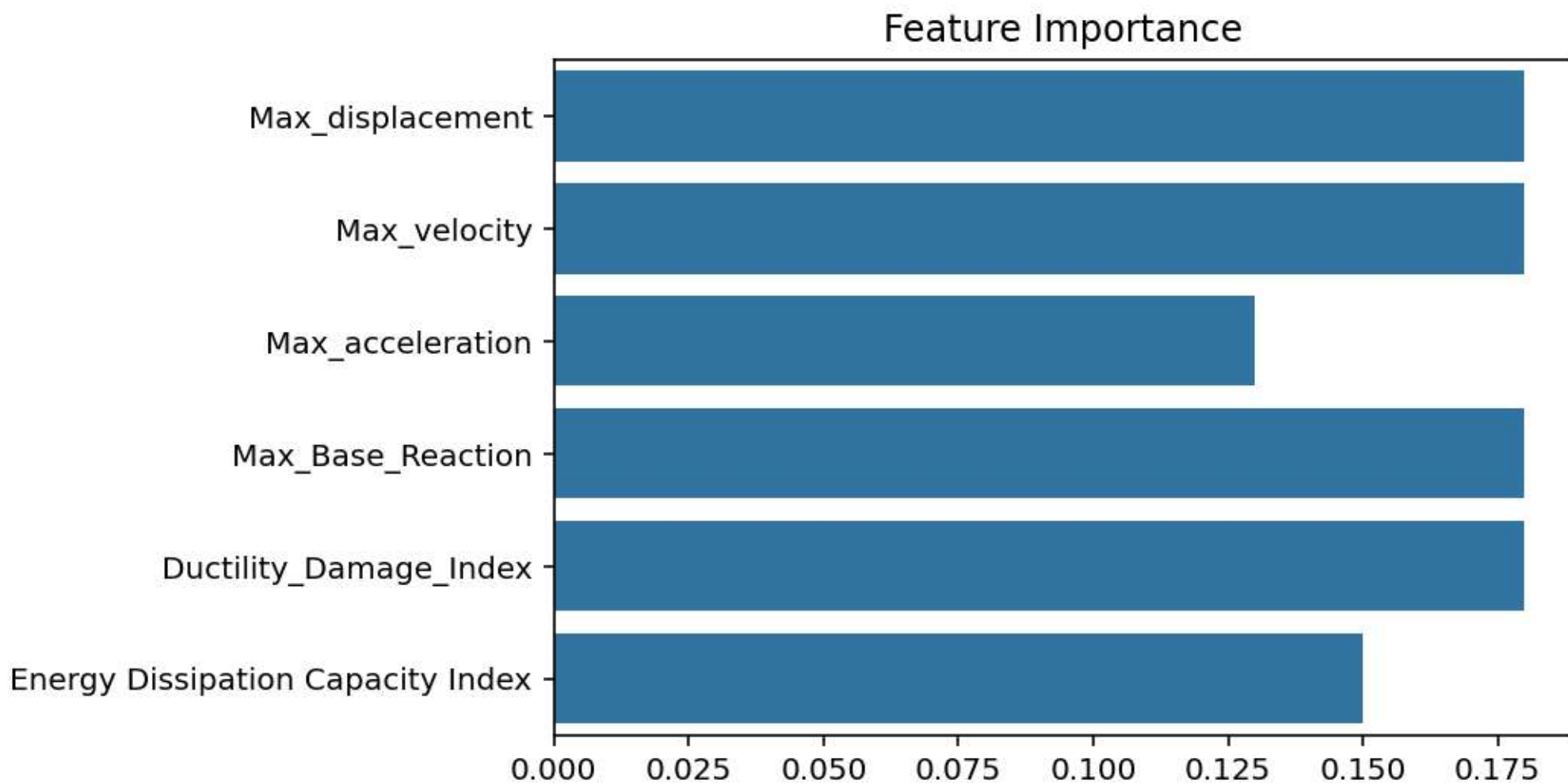








None



Correlation Heatmap

